

WRI INFO FROM WEBSITE:

They say problems are connected,
so their solutions must be connected as well

Their 5-year strategy from 2023 to 2027 includes

- Addressing systemic problems together and viewing problems as interconnected
- Use research-based approaches to work globally to meet people's essential needs
- Protect people, Restore Nature, stabilize the climate, build community
- Moto: Count it. Change it. Scale it
- Technically: produce and use energy, design and manage cities; shift the economic, financial, and governance structures that shape people’s decisions and behaviors

\/ FROM THE BOARD \/

- **Bedrock** -> LLM interface
- **Bedrock AgentCore** -> Agent hosting
- **Bedrock Knowledge Bases** -> Retrieval Augmented Generation (RAG)
- **Amazon S3** -> PDF / Doc. Storage
    - **RDS/Aurora** -> pgvector-based RAG
- Format of Underlying Data
    - Citations

- UI/Frontend
- Data Access/Security
- Types of questions from users
- PII protection
- Data ingestion (+RAG pipeline)

Features In Scope (MVP)

Feature	Description	Priority	Rationale for Inclusion
Corpus ingestion	Load 50 to 150 public WRI documents with basic metadata (provided at event)	Must Have	Required for everything else
Cited Q&A	Answers grounded only in the corpus	Must Have	Core value
"No answer" response	Flag or decline weakly supported answers	Must Have	Limits hallucination
Research briefing	Structured, cited summary of a topic	Must Have	Addresses landscape mapping
Multi-agent workflow	Two or more cooperating agents	Optional	Tests the multi-agent approach
Translation	Translate passages or briefings on request	Should Have	Supports global offices
Research suggestions	Suggest follow-up questions or gaps	Should Have	Differentiator
Simple UI	Chat or web interface for the demo	Must Have	User experience
Figure and table parsing	Complex document context parsed into the knowledge base	Should Have	Many useful documents are complex

Feature goals:
Ease of use for users
Voice recognition
Translation features
Cited sources

AskWRI - World Resources Institute :
https://github.com/12turnbullb/us-breaking-barriers-2026/blob/master/World%20Resources%20Institute/AskWRI-technical-document.md

https://builder.aws.com/

https://builder.aws.com/connect/spaces

Pain points WRI states

1. **Finding past research is slow and incomplete.** Researchers use site search, a database that only covers **2019 onward**, or they ask senior staff. Older publications, especially ones without DOIs, get missed.
2. **They can't see what WRI has already covered on a topic.** There's no easy "map" of past work, so it's hard to know where WRI is strong or thin.
3. **People spend time searching instead of analyzing.** This hits junior and senior staff both, and senior staff get interrupted as the human search engine.
4. **Research in other languages is hard to use.** Global offices need access without relying on someone to translate.
5. **Generic chatbots aren't trusted.** Their answers can't be checked against WRI's own sources, so they're unusable for credible research work.

Splitting the work for 5 people

1. **Data and ingestion:** corpus, S3, Knowledge Base, metadata.
2. **Retrieval and eval:** search quality and the 15 to 25 test questions with scores.
3. **Agents:** Q&A, briefing, and the auditor that checks each sentence.
4. **UI and demo:** the interface, citation highlighting, and the demo script.
5. **Product and story:** talks to WRI, owns the pain points, builds the slides, and runs the extra feature (timeline, translation or coverage map).

Why each current method falls short

- **Site search.** Keyword search on an organization's own website typically matches titles and metadata, not the content inside documents. If a researcher doesn't know the exact terminology used in a 2008 report, they won't find it. PDFs are often poorly indexed, and results aren't ranked by relevance to a research question.
- **A database covering only 2019 onward.** This creates a hard cutoff. Anything earlier is invisible to the tool people actually use, so the effective institutional memory is only a few years deep, even if the organization has decades of work.
- **Asking senior staff.** This works only as long as those people are available, remember accurately, and have time to answer. It doesn't scale, it creates dependency on a few individuals, and it disappears entirely when they retire or leave.

Why older publications without DOIs are hit hardest

A DOI is a persistent identifier that makes a publication discoverable through citation indexes, aggregators, and cross-referencing tools. Publications without one sit outside that infrastructure. They may exist only as scanned PDFs, print archives, or pages on a legacy website, with inconsistent metadata and no links from newer work. So even good tools that rely on DOIs, such as reference managers and citation graphs, can't see them.

The consequences

- **Duplicated work.** Teams re-run studies or re-answer questions the organization already addressed.
- **Missed context.** New research ignores earlier findings, methods, or data that would have strengthened or challenged it.
- **Inconsistent citations.** Without a way to locate prior work, researchers cite only what they can easily find, which skews toward recent material.
- **Uneven access.** New hires and junior staff are disadvantaged compared with those who know whom to ask.
- **Lost institutional knowledge.** Over time, older work is effectively forgotten even though it remains valuable.

Ali's Kiro Generated Plan:

Name suggestions:

- Nexora
- Syntara
- Orbis
- Auralis

6 ideas for features:

1. Research Connection Map

My top pick

Instead of returning just documents, AskWRI shows how WRI research connects across topics.

Example: A researcher searches "Deforestation in Brazil." The system connects findings about deforestation, rainfall, agriculture, and water scarcity.

Why valuable: Directly aligns with WRI's strategy that environmental problems are interconnected.